

INFLUENCE OF PREACTIVITY AND ECCENTRIC MUSCLE ACTIVITY ON CONCENTRIC PERFORMANCE DURING VERTICAL JUMPING

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ABSTRACT

The purpose of this investigation was to observe the influence of increasing amounts of preactivity and eccentric muscle activity imposed by three different jump types on concentric vertical jumping performance. Sixteen athletes involved in jumping-related sports at Appalachian State University, which is a Division IA school, performed a static jump (SJ), counter-movement jump (CMJ), and drop jump (DJ). Force, power, velocity, and jump height were measured during each jump type. In addition, muscle activity was measured from two agonist muscles (vastus lateralis, vastus medialis) and one antagonist muscle (biceps femoris). Preactivity and eccentric phase muscle activity of the agonist muscles (average integrated electromyography) was significantly ($p \leq 0.05$) higher during the DJ (preactivity, 0.2 ± 0.11 mV; eccentric phase, 1.00 ± 0.36 mV) in comparison with the CMJ (preactivity, 0.11 ± 0.10 mV; eccentric phase, 0.45 ± 0.17 mV). Peak concentric force was highest during the DJ and was significantly different among all three jump types (SJ, CMJ, DJ). Maximal jump height was significantly higher during the DJ (0.41 ± 0.05 m) and CMJ (0.40 ± 0.06 m) compared with the SJ (0.37 ± 0.07 m). However, no significant difference in jump height existed between the CMJ and DJ. A positive energy balance, as assessed by force-displacement curves during the eccentric and concentric phases, was observed during the CMJ, and a negative energy balance was observed during the DJ. The data from this investigation indicate that a significant increase in concentric vertical jump performance is associated with increased levels of preactivity and eccentric phase muscle activity (SJ to CMJ). However, higher eccentric loading (CMJ to DJ) leads to a negative energy balance during the eccentric phase, which may relate to a non-significant increase in vertical jump height, even with coincidental increases in

peak concentric force. Practitioners may want to focus on improving eccentric phase muscle activity through the use of plyometrics to improve overall jumping performance in athletes.

KEY WORDS stored elastic energy, stretch-shortening cycle, drop jump

INTRODUCTION

The purpose of the current investigation was to determine whether increasing the amount of preactivity and eccentric phase muscle activity during vertical jumping would improve concentric phase performance. To address this issue, a comparison of a static jump (SJ), counter-movement jump (CMJ), and drop jump (DJ) was performed. It has been previously reported that these jumping patterns result in an increasing amount of pre-activation and eccentric phase muscle activity from a SJ to a CMJ to a DJ (10). If increasing pre-activation and eccentric phase muscle activity has a positive influence on concentric work, then it is theorized that concentric variables related to jumping performance should improve accordingly. However, based on a previous investigation, the observation of a positive or negative energy balance that may occur during each jump type may also influence concentric performance (7).

In studies that have compared a SJ and CMJ, jump height has been significantly higher during a CMJ (8,10,12). The active state of the muscle may be increased during the preparatory or eccentric phase of the CMJ and allow for more work to be performed concentrically (2). A higher level of eccentric muscle activity has been observed in the CMJ with no significant changes in the level of concentric muscle activity between a SJ and CMJ (12). However, when comparing a DJ to CMJ, the associated increase in eccentric muscle activity during a DJ does not seem to translate into increased jump height (10,12). This may be because of the positive energy balance observed in a CMJ as opposed to the negative energy balance associated with a DJ (10). The CMJ force-displacement curve indicates the level of energy absorbed during the eccentric phase is lower than that of the energy produced during the concentric phase, with a respective DJ showing an opposing pattern (10).

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22(3)/750-757

Journal of Strength and Conditioning Research
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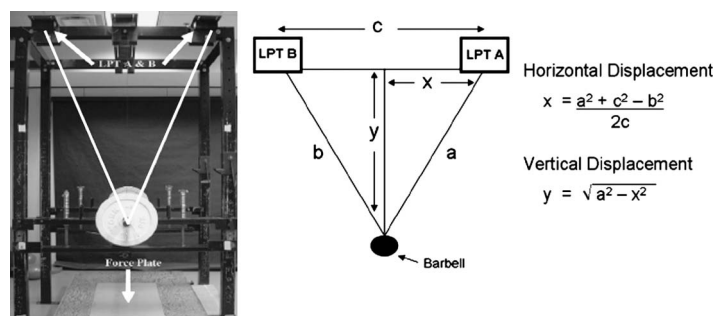


Figure 1. Custom-built power rack with linear position transducers (LPT) used to measure horizontal and vertical bar displacement.

Although hypothesized to be true, these issues have never been thoroughly evaluated utilizing examination of a large prime mover (quadriceps) among three vertical jump types (SJ, CMJ, DJ) simultaneously. The purpose of this investigation was to make a direct comparison of the levels of muscle preactivity and eccentric phase muscle activity and their influence on concentric performance. Three primary hypotheses were addressed: 1) increasing large prime mover

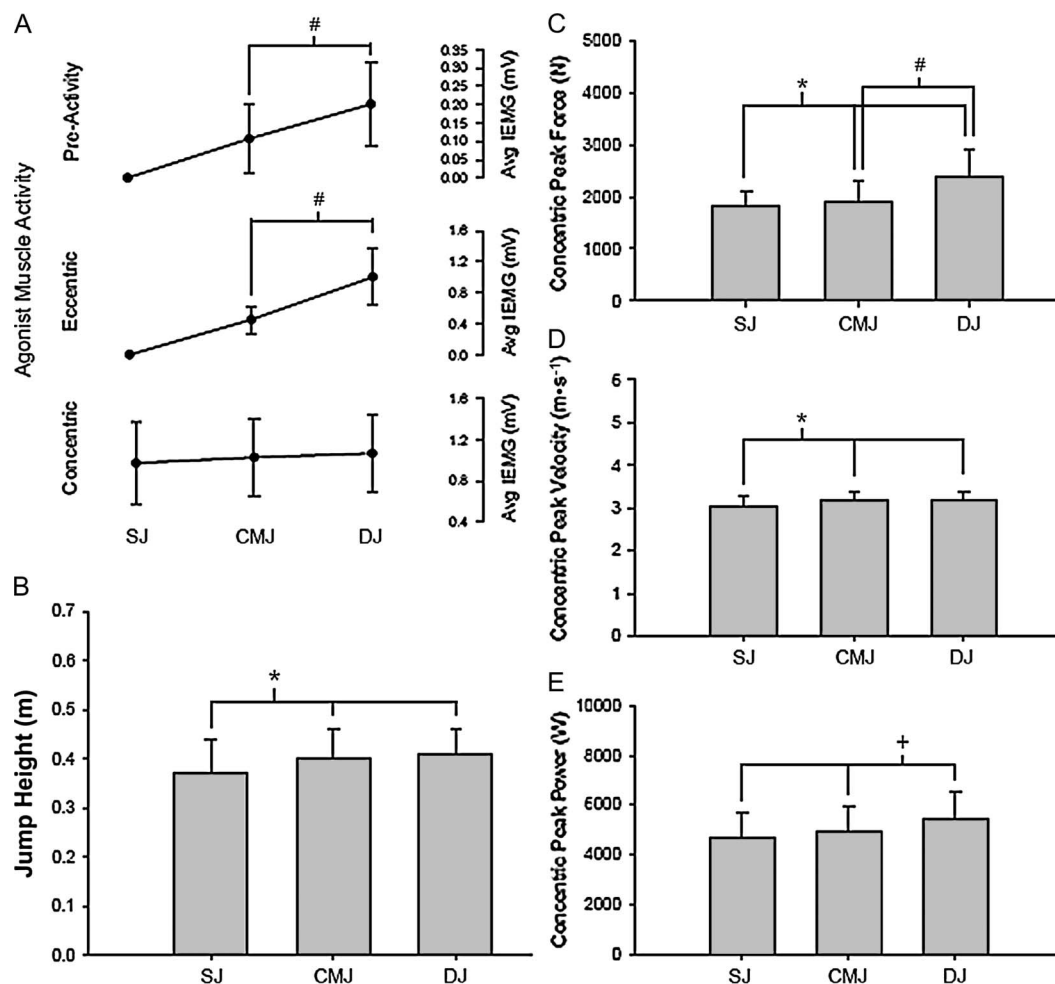


Figure 2. (A) Mean $\pm$ SD for average integrated electromyography (Avg IEMG) of pre-activity and eccentric and concentric phases of the different jump conditions, static jump (SJ), countermovement jump (CMJ), drop jump (DJ) for agonist muscles vastus lateralis (VL) and vastus medialis (VM). (B) Maximal jump height in meters. (C) Peak force in Newtons. (D) Peak velocity in meters per second. (E) Peak power in Watts. #Significant difference ($p \leq 0.05$) between CMJ and DJ. *Significant difference ($p \leq 0.05$) from CMJ and DJ compared with SJ. +Significant difference ($p \leq 0.05$) from SJ and CMJ compared with DJ.

TABLE 1. Average integrated EMG during each jump condition.

	Agonist muscles			Antagonist muscles		
	Pre-activity	Eccentric	Concentric	Pre-activity	Eccentric	Concentric
SJ			0.97 ± 0.40			0.15 ± 0.09
CMJ	0.11 ± 0.10	0.45 ± 0.17	1.03 ± 0.37	0.02 ± 0.02	0.08 ± 0.05	0.16 ± 0.08
DJ	0.20 ± 0.11*	1.00 ± 0.36*	1.07 ± 0.37	0.08 ± 0.07*	0.13 ± 0.13	0.15 ± 0.08

Values are mean ± SD expressed in millivolts.

EMG = electromyograph; SJ = static jump; CMJ = counter-movement jump; DJ = drop jump.

*Significant ($p \leq 0.05$) difference in muscle activity between the CMJ and DJ.

agonist muscle preactivity and eccentric phase muscle activity (SJ to CMJ) would significantly improve concentric vertical jumping height, 2) increasing large prime mover agonist muscle preactivity and eccentric phase muscle activity would increase force output during the concentric phase (SJ to CMJ to DJ), and 3) increasing pre-load above a certain level (CMJ to DJ) would lead to a negative energy balance when analyzing a large prime mover and thus lead to no significant improvement in vertical jump height. Only two other investigations have attempted to make this comparison. The first investigation in this area utilized a very small sample size ($n = 6$) (23). In the second investigation, analysis was performed on the triceps surae muscle complex, in contrast to the current study, which examined the quadriceps muscle (10). Eccentric-concentric muscle activity patterns and their contribution to increased concentric work have been shown to be different among various muscle groups (1). Thus, this investigation attempted to establish the relationship between the muscle patterns of the major agonist (quadriceps) in vertical jumping and their influence on concentric performance with various levels of muscle preactivity and eccentric phase muscle activity in the context of a SJ, CMJ, and DJ.

METHODS

Experimental Approach to the Problem

The subjects completed one testing session. Force, power, velocity, jump height, and electromyography (EMG) from the vastus lateralis (VL), vastus medialis (VM), and biceps femoris (BF) were recorded during all jumps. All subjects warmed up

on a bicycle ergometer for 5 minutes at an intensity of 1 kilopond before the vertical jump testing. The athletes were instructed to perform the vertical jumps with maximal effort while striving for maximal vertical displacement in each trial. Five trials were allowed for each vertical jump type with the best trial used for comparison. Each jump was performed as described: i) SJ performed at a starting knee angle of 100° with no countermovement to a maximal jump height; ii) CMJ performed from a standing position to a 100° knee angle rebounding to a maximal jump height; and iii) DJ performed from a drop height equivalent to the maximal jump height attained during the CMJ rebounding to a maximal jump height. The jump procedures and knee angles utilized are based on previous investigations in our laboratory (4,17,21). The order in which the jump types were performed was randomized. The DJ height was made relative to the subject's maximal CMJ height to account for differences in strength

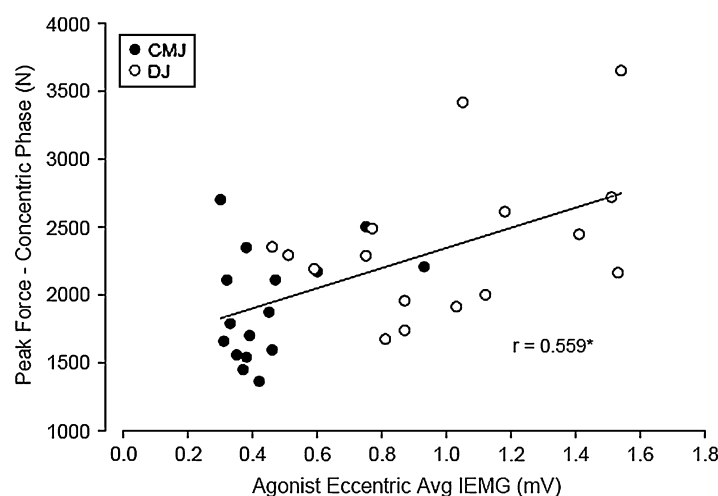


Figure 3. Relationship between agonist (vastus lateralis (VL) and vastus medialis (VM)) eccentric muscle activity (average integrated electromyography [Avg IEMG]) and the resultant peak concentric vertical force (F_z) during the countermovement jump (CMJ) and drop jump (DJ). *Significant correlation ($p \leq 0.05$).

levels and jumping ability within the subjects. Furthermore, the DJ height was chosen to elicit significant amounts of eccentric preload and to mimic the rebounding that may occur in the athletes' specific sports when repeated jumps are attempted (16).

Subjects

Eight male and eight female National Collegiate Athletic Association Division IA basketball and volleyball players ((mean $\pm$ standard deviation) height 180 ± 7 cm; weight 80 ± 7 kg; body fat $15\% \pm 4\%$) were recruited through their respective coaches to participate in the current study. The male and female data were analyzed separately for differences in kinetic, kinematic, and EMG variables. No gender differences were found (with respect to differences between the SJ, CMJ, and DJ); therefore, the data were analyzed together, and it was assumed that gender differences did not exist in this sample. The athletes were involved in in-season or pre-season training and visited the laboratory on a day off from their normal sport training to participate in the study. The subjects were chosen because their sports require repetitive maximal effort jumping in practice and competition, and they were considered to be trained jumpers. The volunteers were notified about the potential risks involved and gave their written informed consent; this research was approved by Appalachian State University's institutional review board.

Measurement of Kinetic and Kinematic Variables

All testing was performed with the subjects standing on a force plate (BP6001200; AMTI, Watertown, Mass.) with the left and right side of a zero load bar attached to two linear position transducers (LPT) (PT5A-150; Celesco Transducer Products, Chatsworth, Calif.) (Figure 1). The data from the right side of the bar were used for analysis. Analog signals from the force plate and two LPTs (right side of the bar) were collected for every trial at 1000 Hz using a BNC-2010 interface box with an analog-to-digital card (NI PCI-6014; National Instruments, Austin, Tx.). Custom software designed in LabVIEW (National Instruments) was used for recording and analyzing the

data. Signals from the two LPTs and the force plate, as well as data derived using double differentiation, underwent rectangular smoothing with a moving average half-width of 12. From laboratory calibrations, the LPT and force plate voltage outputs were converted into displacement and vertical ground reaction forces, respectively. Displacement and velocity from every repetition within each jump type (SJ, CMJ, DJ) were calculated utilizing the two LPTs, which allowed for both vertical and horizontal movements to be measured. The LPTs were mounted above and anterior (LPT A), and above and posterior (LPT B) the subject, forming a triangle when attached to the barbell (Figure 1). Combining known displacements with the displacement measurements from LPT A and B in the equation defined in Figure 1 allowed for the calculation of vertical displacement, simultaneously accounting for any horizontal movement of the bar. Power-time curves were generated by multiplying the force-time curve and velocity-time curve. Reliability of this method was previously assessed through comparison of peak power between two trials during a CMJ, a traditional

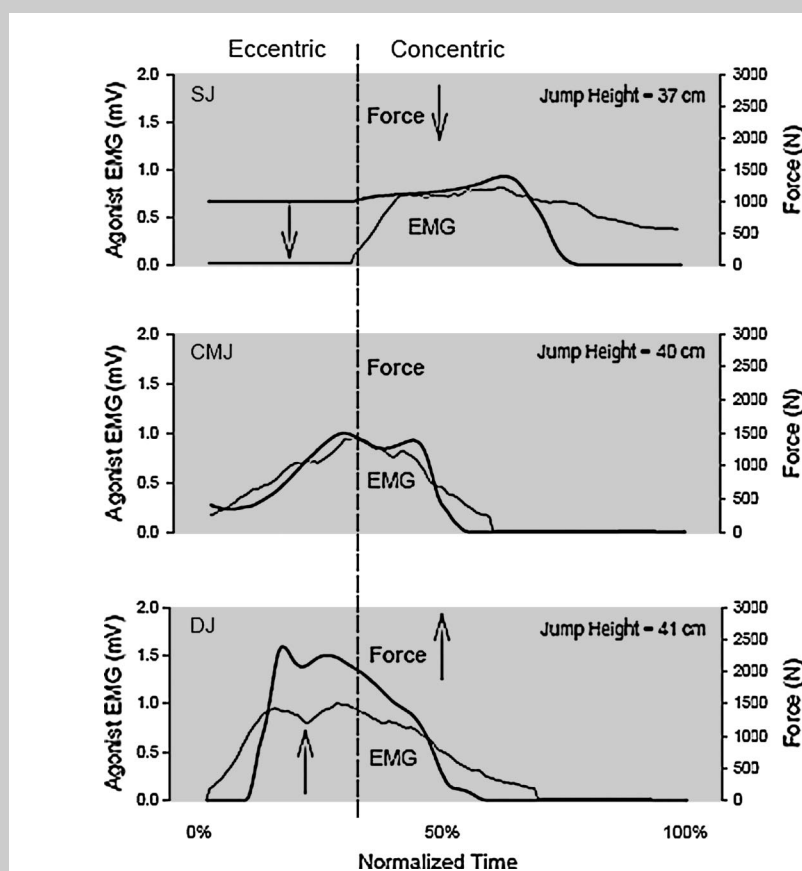


Figure 4. Force-normalized time and electromyography-normalized time curves for the eccentric and concentric phase of the static jump (SJ), counter-movement jump (CMJ), and drop jump (DJ). Arrows indicate no eccentric phase muscle activity for SJ and increased eccentric phase muscle activity for the DJ compared with the CMJ. Arrows indicate decreased force during the concentric phase of SJ and increased force during the DJ compared with the CMJ. Peak vertical jump height indicated for reference purposes.

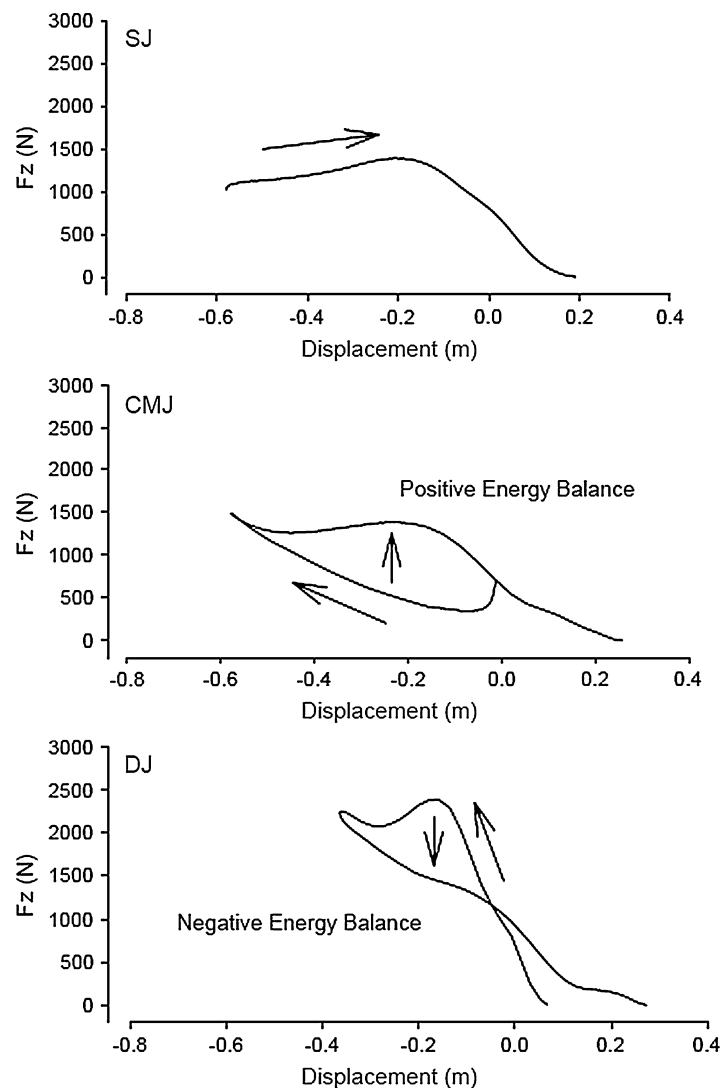


Figure 5. Average force-displacement or work loops of the static jump (SJ), counter-movement jump (CMJ), and drop jump (DJ). Arrows indicate the progression of the jump from the eccentric to concentric phase. Negative versus positive energy balance represents the greater amount of force and work occurring in the eccentric phase of a DJ versus a CMJ.

squat (S), and a power clean (PC). The method displayed intraclass correlation coefficients (ICC) above the minimal acceptable criterion of 0.70 and were significant at an α level of 0.05 (CMJ, 0.95; S, 0.94; PC, 0.98). This method of power calculation was also previously validated in comparison to the calculation of power using only a force plate. The peak power of 10 subjects was not significantly different between the two methods during the CMJ at loads ranging from 0% to 85% of one repetition maximum (1RM) (2 LPTs + force plate 0% of 1RM peak power = 6332 ± 1085 ; 2 LPTs + force plate 85% of 1RM peak power = 3986 ± 563 ; force plate only 0% of 1RM peak power = 6260 ± 1181 ; force plate only 85% of 1RM peak power = 4247 ± 868).

Electromyography

EMG activity from the VM, VL, and BF of the subject's dominant jumping leg was recorded during all of the jumps using a telemetry transmitter (eight-channel, 12-bit analog to digital converter; Noraxon USA Inc., Scottsdale, AZ) as described previously (19,20). A disposable surface electrode (2-cm inter-electrode distance, 1-cm circular conductive area; Noraxon USA Inc.) was attached to the skin over the belly of each measured muscle, distal to the motor point, and parallel to the direction of muscle fibers. The amplified myoelectric signal recorded during the vertical jumps was detected by the receiver-amplifier (Telemetry 900, gain = 2000, differential input impedance = 10 M Ω , bandwidth frequency 10–500 Hz, common mode rejection ratio = 85 dB, Noraxon USA) and then sent to an A/D card at 1,000 Hz. Custom-designed software created in LabVIEW was used for recording and analyzing the data. The signal was full wave-rectified and filtered (six-pole Butterworth, notch filter 60 Hz, band pass filter 10–200 Hz). The integrated value (mV·s⁻¹) was calculated and averaged over the determined phase (preactivity, eccentric, concentric) of each vertical jump to determine average integrated EMG (Avg IEMG) (mV). Pre-

activation for the CMJ was defined as 100 ms before the vertical force (Fz) unloading phase. Pre-activation for the DJ was defined as 100 ms before foot contact (15,16). The end of the eccentric phase was defined as the time point at which vertical bar velocity became positive (16). Pre- and eccentric phase muscle activity was not reported during a SJ because this portion of the jump does not contribute to concentric performance (14,16). The end of the concentric phase was defined as the point at which Fz reached 0.

Statistical Analyses

A general linear model multivariate analysis with Bonferroni post hoc tests was used to identify within- and between-group

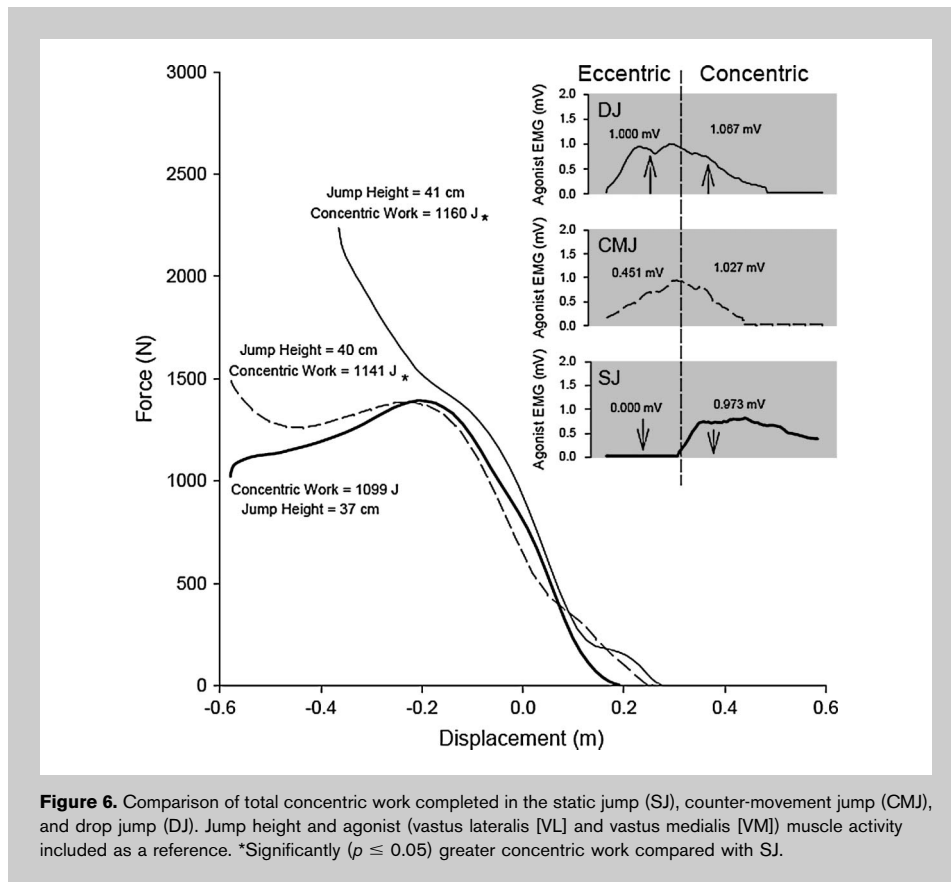


Figure 6. Comparison of total concentric work completed in the static jump (SJ), counter-movement jump (CMJ), and drop jump (DJ). Jump height and agonist (vastus lateralis [VL] and vastus medialis [VM]) muscle activity included as a reference. *Significantly ($p \leq 0.05$) greater concentric work compared with SJ.

differences. The criterion α level was set at $p \leq 0.05$. All statistical analyses were performed through the use of a statistical software package (SPSS Version 11.0; SPSS Inc., Chicago, IL).

RESULTS

Muscle Activity

Figure 2A and Table 1 display agonist (VM and VL) muscle activity (Avg IEMG) for preactivity and the eccentric and concentric phases during the SJ, CMJ, and DJ. Preactivity was significantly higher during the DJ compared with the CMJ. Eccentric phase muscle activity was also significantly higher during the DJ compared with the CMJ. Table 1 also presents Avg IEMG data for the antagonist (BF) muscles for preactivity and the eccentric and concentric phases. Antagonist muscle preactivity was significantly higher during the DJ compared with the CMJ. Concentric phase muscle activity was not significantly different among the three types of jumps.

Concentric Performance

Maximal jump height was significantly lower during the SJ compared with both the CMJ and DJ (Figure 2B). Maximal jump height was not significantly different between the CMJ and DJ. Figure 2C–E illustrate peak concentric Fz, peak concentric vertical bar velocity, and peak concentric vertical

power, respectively. Peak concentric Fz was significantly different among all three types of vertical jumps (SJ, CMJ, DJ). Peak concentric velocity was significantly lower during the SJ compared with the CMJ and DJ. Peak concentric velocity was not significantly different between the CMJ and DJ. Peak concentric power was significantly higher during the DJ compared with the CMJ and SJ. Peak concentric power was not significantly different between the CMJ and SJ.

Force and Muscle Activity

Peak concentric Fz and eccentric Avg IEMG of the agonist muscles (VL and VM) were significantly correlated ($r = 0.559$, power at α level of 0.05 = 0.925) (Figure 3). The force-time curves and agonist muscle activity curves averaged for all 16 subjects for the SJ, CMJ, and DJ are displayed in Figure 4. Jump height was also reported in this Figure for reference purposes. Note the increased

eccentric phase muscle activity and increased concentric phase force output during the DJ versus the SJ and CMJ.

Energy Balance and Work

Figure 5 shows the force-displacement or work loops for the three types of jumps (SJ, CMJ, DJ). Notice the positive energy balance that exists for the CMJ and the negative energy balance during the DJ. The amount of concentric work performed with respect to a given amount of agonist concentric muscle activity among the SJ, CMJ, and DJ is represented in Figure 6. Concentric work was significantly greater in the CMJ and DJ compared with the SJ, with no differences present between CMJ and DJ. EMG time curves are presented for reference purposes.

DISCUSSION

The data from the current investigation support the theory that increasing large prime mover agonist muscle preactivity and eccentric phase muscle activity (SJ to CMJ) significantly improves vertical jumping height. Increasing prime mover agonist muscle preactivity and eccentric phase muscle activity increases force output during the concentric phase (SJ to CMJ to DJ). In addition, increasing pre-load above a certain level (CMJ to DJ) leads to a negative energy balance when examining a large prime mover (quadriceps) and thus leads to no significant improvement in vertical jump height. Finally,

no differences in concentric muscle activity during the SJ to CMJ to DJ were observed for any of the muscles examined. The patterns observed were independent of gender; thus, the data are presented as a combination of both male and female subjects.

When examining the data from the SJ and CMJ in the current investigation, the level of agonist concentric muscle activity was not significantly different. However, there was a significant difference in jump height, peak concentric force, and peak concentric velocity. Considering that no eccentric muscle activity was present in the SJ, the increased concentric performance during the CMJ must be a result of muscle activity during the eccentric phase. The mechanism referred to above indicating the ability of the muscle tendon unit to store and release elastic energy (16) seems plausible but was not provable in the current investigation. Recent animal model data provide convincing evidence that stored elastic energy is a significant contributor to vertical jump performance in a CMJ (13).

The timing for release of stored elastic energy from the tendon may be optimized with an increasing amount of eccentric muscle activity (8). This means that with increasing eccentric phase muscle activity, the time of release of stored elastic energy from the tendon may move to later in the concentric phase. Figure 6 provides support that this, in fact, may be true, as the amount of work performed during the concentric phase of the DJ extended well into the concentric phase. In comparison, the increased work during the concentric phase for the CMJ was limited to the very early portions of the concentric phase. The current data are similar to previous findings in that the time frame for increased work performed during the concentric phase as a result of a subsequent eccentric muscle action during a CMJ activity was relatively short-lived (6). It seems that the contribution of an eccentric phase to extending work throughout the entire length of the concentric phase may be enhanced by increasing pre-load (DJ). However, this is speculative at best and did not lead to any appreciable increase in vertical jump height during the DJ in the current investigation.

Increasing eccentric phase muscle activity seems to increase force output during the concentric phase. The data in Figure 3 support this theory. The force output capabilities of a lengthened active muscle are clearly increased (3). The data from the current investigation cannot confirm the existence of the observed *in vitro* phenomenon but lead to further speculation as to the possibility of these physiological mechanisms influencing concentric muscle force output during a CMJ. It is unclear why the increased force production during the concentric phase of the DJ did not lead to a significant increase in vertical jump height in comparison to the CMJ. However, the negative energy balance that occurred during the DJ but not during the CMJ may be a possible explanation. This phenomenon has been observed previously (10). Increased muscle stiffness may allow for maintenance of a positive energy balance with

increasing levels of pre-load. This was clearly illustrated in a previous study by Fukashiro et al. (9), who reported that black athletes had significantly greater lower leg stiffness compared with white athletes, and that this was related to jumping and sprinting performance. The current investigation provides no further insight, however, as muscle stiffness and its relationship to maintaining a positive energy balance during the DJ were not assessed.

The present data suggest that muscle activation of VL, VM, and BF during the concentric phase was not a contributing factor to the enhancement of concentric jump performance variables observed in the CMJ and DJ vs. SJ. The absence of a pre-stretch in the SJ condition did not lead to a decrease in concentric muscle activity when compared with CMJ and DJ. These data are in agreement with those of Walshe et al. (24), who reported no differences in concentric muscle activity between a SJ and CMJ. In addition, these observations may be explained by the behavior of the fascicles during the pre-stretch conditions observed during a CMJ and DJ. Ishikawa et al. (16) reported that during a maximal DJ, the fascicle length change decreases and a greater percentage of muscle tendon unit length change occurred in the tendon of the VL. Furthermore, the lengthening velocity of the fascicle was observed to decrease with increased pre-stretch intensity. As the intensity of the pre-stretch increases, muscle stiffness is regulated to allow a greater stretch of the tendon, which increases storage of elastic energy (16).

In conclusion, it seems that increasing muscle preactivity and eccentric phase muscle activity enhances variables related to concentric jumping performance. A CMJ allows for increased jump height, peak concentric force, and peak concentric velocity. However, increasing preactivity and eccentric phase activity above the level of the CMJ (to the DJ) only resulted in increased concentric force output but no significant improvement in jump height. The increased preactivity and eccentric phase muscle activity may increase the mechanical efficiency of the DJ over CMJ through changes in muscle tendon unit stiffness regulation (1). It is suggested that increased levels of eccentric muscle activity are only beneficial for increasing vertical jump height if a positive energy balance can be maintained during the eccentric phase of the jump. Thus, increased muscle stiffness and its ability to allow an individual to maintain a positive energy balance during increasing levels of pre-load should be examined in future research.

PRACTICAL APPLICATIONS

Increasing muscle preactivity and eccentric phase muscle activity seems to result in increased jump height. Plyometric training may be utilized to induce increases in muscle preactivity and thus may be a useful tool in increasing athletic performance (18). Lower-body stiffness regulation may assist in maintaining a positive energy balance during jumping and thus also improve jump performance. Therefore, the role of exercise and stretching and their influence on muscle

stiffness should also be considered by practitioners (5,7,11). In conclusion, practitioners should incorporate plyometric training, which increases eccentric phase muscle activity, and thus should result in increased jumping ability in their athletes.

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